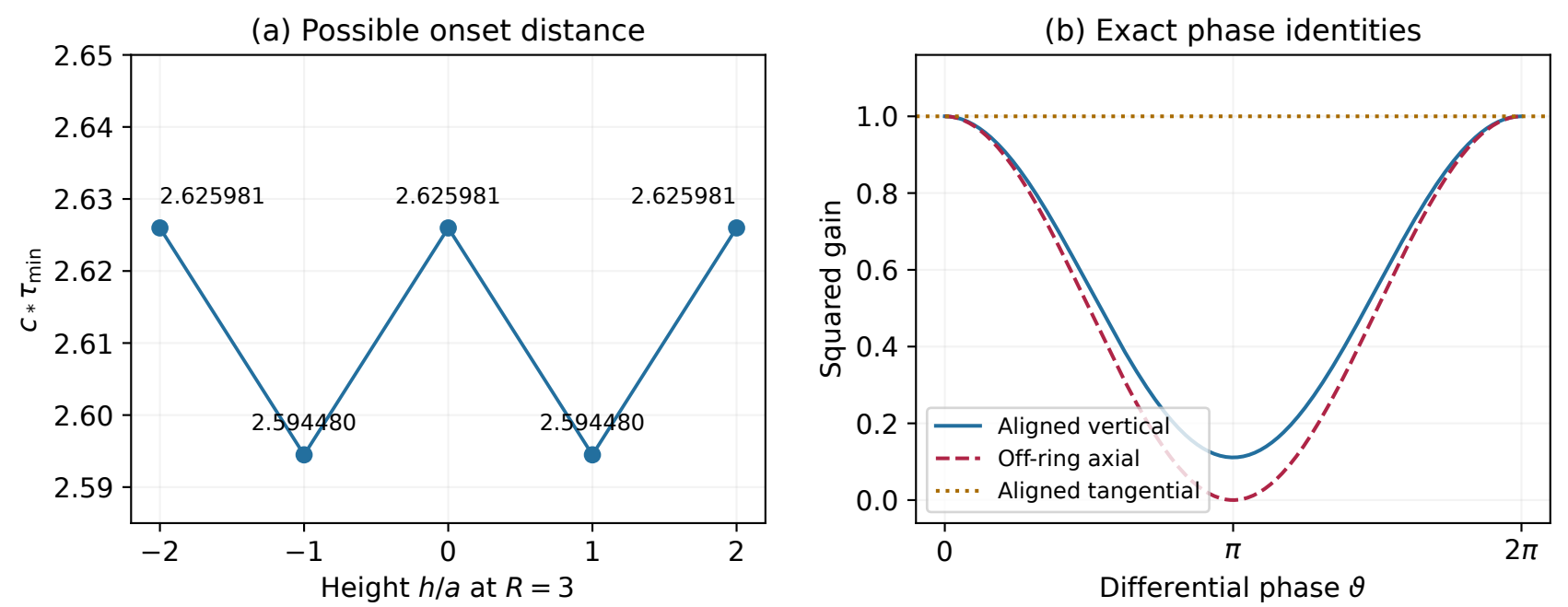
(a) Possible onset distance
2.65
2.64
2.63
2.62
2.61
2.60
2.59
$c * \tau_{min}$
2.625981
2.625981
2.625981
2.594480
2.594480
−2
−1
0
1
2
Height h/a at $R = 3$
(b) Exact phase identities
1.0
0.8
0.6
0.4
0.2
0.0
Squared gain
Aligned vertical
Off-ring axial
Aligned tangential
0
π
2π
Differential phase ϑ